



SWI-650-SS, Assembly SWI, 650 Drilling Shock Sub

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Legal Information

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Document Control

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Revision History

Version	Date	Description	Prepared by
1	11/20/2025	Initial Version	A. Rivera
2	1/15/2026	Added back-up ring orientation to step 2.5.	A. Rivera
3	3/12/2026	Added torque values for the lock nut and set screws (Table 3-1).	A. Rivera

Regulatory Compliance

Waste Management

IMPORTANT INFORMATION FOR CORRECT DISPOSAL OF THE EQUIPMENT



This symbol means that the equipment cannot be discarded in a rubbish-bin. At its end of life, the equipment and/or its components must be treated following Tethys environmental procedures, in compliance with the Tethys QHSE Policy and applicable laws and regulations on waste management.



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1 Introduction

PPE



The SWI-650-SS standard work instruction (SWI) gives the procedure for the assembly of the 650 drilling shock sub.

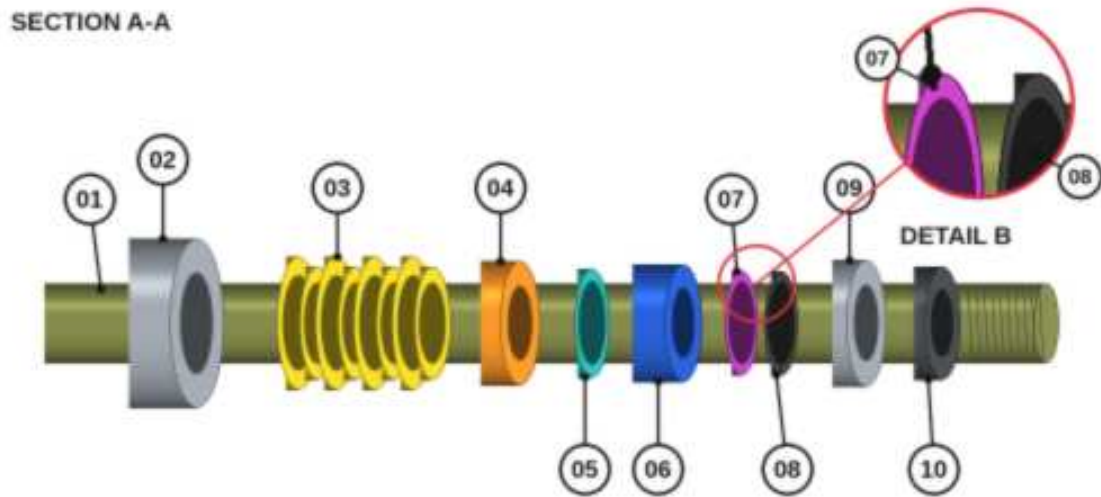
Obey all safety requirements as given in the Tethys Safety Standards. Important safety requirements are:

- Obey PPE requirements as per the Tools and Equipment PPE Appendix to Tethys QHSE Standard S-003.
- Use safe lifting methods when you move parts and tooling around the work area.
- Make sure that only necessary personnel are in the work area.
- Make sure that only approved and certified personnel use machinery during the procedure.
- Obey all safety requirements for electrical equipment.

Discard any equipment, component or consumables in accordance with the Tethys QHSE policy and applicable laws and regulations on waste management.

Parts List

Item	Part No.	Description	Qty
01	TDT-65001	Mandrel, 6-1/2	1
02	TDT-65002	Housing, 6-1/2	1
03	TDT-65003	Disc spring, 4.25 OD	6
04	TDT-65004	Spline sleeve	1
05	TDT-65005	Wiper ring	1
06	TDT-65006	Seal carrier	1
07	TDT-65007	O-ring, 2-348, HNBR 90	2
08	TDT-65008	Back-up ring, PEEK	2
09	TDT-65009	Thrust bearing	1
10	TDT-65010	Lock nut, 4-1/2 UN	1
11	TDT-65011	Set screw, 3/8-16 x 1/2	2
12	TDT-65012	Grease fitting, 1/8 NPT	1



Mandrel stack, exploded view

Figure 1-1 Mandrel stack components

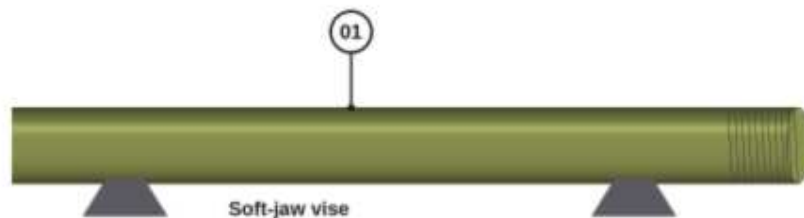
2 Assemble the Mandrel Stack

PPE required



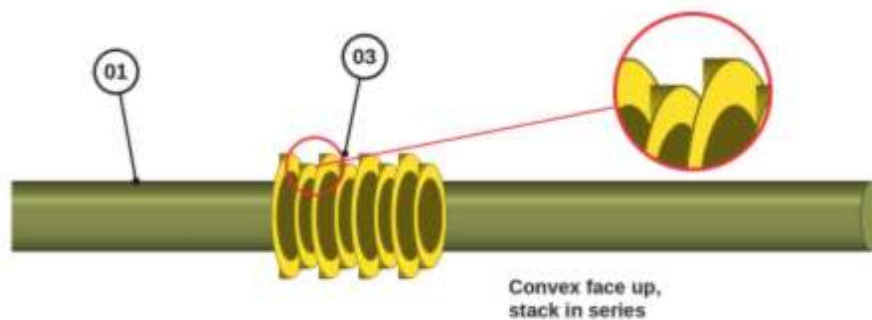
! Make sure that all components are clean before you start the assembly.

2.1 Hold the mandrel (01) horizontally in a soft-jaw vise.



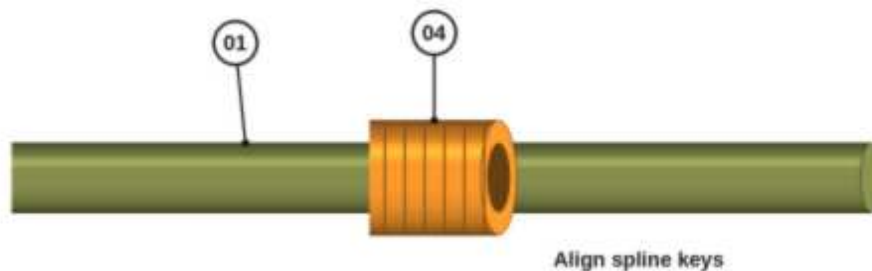
i Make sure that the vise jaws do not damage the chrome surface of the mandrel.

2.2 Install eight new disc springs (03) onto the mandrel (01).



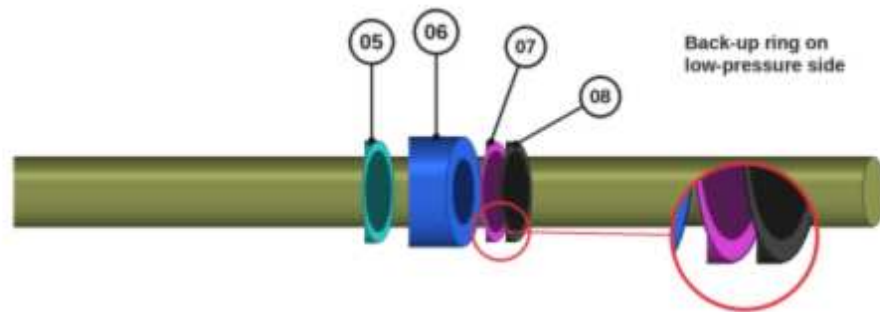
i Stack the disc springs in series with the convex face up. Springs installed in the wrong direction will fail under load.

2.3 Install the spline sleeve (04) onto the mandrel (01). The spline keys must align with the slots on the mandrel.



2.4 Install the wiper ring (05) and the seal carrier (06) onto the mandrel (01).

- 2.5** Install the O-ring (07) and the back-up ring (08) into the groove of the seal carrier (06). The back-up ring must be on the low-pressure side.



- 2.6** Install the thrust bearing (09) against the seal carrier (06).

3 Install the Mandrel Stack into the Housing

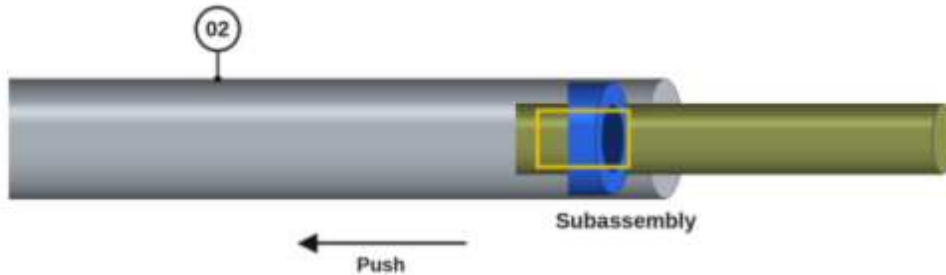
PPE required



- 3.1** Secure the housing (02) to prevent movement in any direction. Examine the inside of the housing for debris. Clean if necessary.

! Make sure that all components are clean before you start the assembly.

- 3.2** Use a sling and overhead crane to install the subassembly into the housing (02). Push it as far forward as possible.



i Use V-blocks, jack stands or other methods per local best practice.

- 3.3** Install the lock nut (10) and torque it to TQ-B (see Table 3-2).



- 3.4** Install two set screws (11) into the lock nut (10) and torque them to TQ-A.

Table 3-1 Torque Values

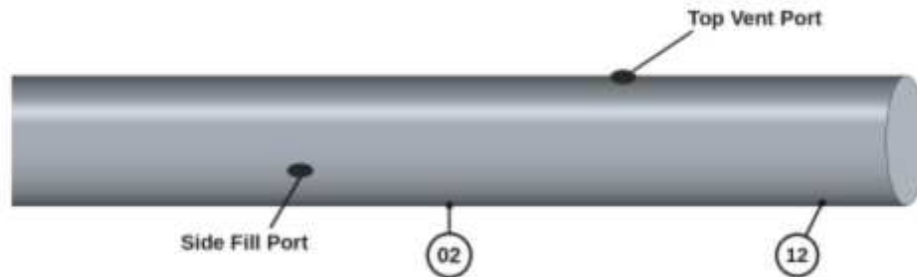
Code	Connection	Torque (ft-lbf)	Tool
TQ-A	Set screw (11) to lock nut (10)	35 - 40	Hex bit, 3/16
TQ-B	Lock nut (10) to mandrel (01)	1,800 - 2,000	Spanner wrench, 650

4 Grease Fill

PPE required



- 4.1 Remove the plugs from the side fill port and the top vent port of the housing (02).



- 4.2 Pump grease through the side fill port until clean grease flows from the top vent port without air bubbles.

! Do not exceed 50 psi grease pressure. Excess pressure can push the seal carrier (06) out of position.

- 4.3 Install the grease fitting (12) and the vent plug.

- 4.4 Record the grease volume on the assembly checklist.